Smart Computer Health Monitor - Diagnostic Report

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Domain: Operating Systems & Resource Health Interpretation

1. Executive Summary & Resource Aggregates

Metric	Value
Total Metric Samples	96
Average CPU Utilization	46.9%
Peak CPU Utilization	88.7%
Average RAM Utilization	90.7%
Peak RAM Utilization	99.5%
Average Health Score	61.6 / 100
Minimum Health Score	47.6 / 100

2. Explainable Health Insights (Interpretation Layer)

Severity / Resource	Diagnosis & Evidence	Possible Cause & Action
[CRITICAL] RAM	**High Memory Pressure Detected** RAM utilization is at 88.6%, leaving only 1858.9 MB available out of 16251.8 MB. When physical RAM is depleted, the operating system pages inactive memory blocks to disk swap/pagefile, which can significantly degrade system responsiveness. *Observed: 88.6% used (1858.9 MB free)*	**Possible factor:** Processes consuming large working sets (vmmemWSL (1864.5 MB), MsMpEng.exe (361.5 MB), python.exe (212.9 MB)) may have extensive data loaded, such as multiple browser tabs, virtual machines, or caching services. High RAM usage does not by itself prove a memory leak. **Action:** Review open browser tabs, background IDEs, or large datasets. Close memory-intensive applications that are not currently in use.
[CRITICAL] Disk	**Critical Storage Exhaustion** Target storage volume (C:\) is 91.7% full, with only 19.7 GB remaining. Operating systems require free disk space for temporary files, virtual memory paging, system updates, and application crash dumps. *Observed: 91.7% (19.7 GB free)*	**Possible factor:** Accumulated software caches, downloads, system log archives, or large media files on the drive. Disk capacity exhaustion does not indicate hardware drive failure. **Action:** Run Disk Cleanup or inspect user download folders and temporary caches. Relocate large inactive media or archive files to external or secondary storage.

3. Actionable Advisory Recommendations

Priority	Recommendation	Expected Benefit & Safety Note

High	**Relieve Working-Set Memory Pressure (vmmemWSL, MsMpEng.exe)** Physical RAM allocation is high (88.6% used (1858.9 MB free)), increasing risk of swap/paging thrashing.	Prevents disk swap paging thrash, keeping foreground applications responsive. *Safety: Ensure any unsaved documents or browser form submissions are saved prior to closing applications.*
High	**Reclaim Free Storage Capacity on C:** Storage volume has only 19.7 GB remaining (91.7% (19.7 GB free)). Operating systems need free space for updates and paging.	Ensures sufficient storage margin for operating system updates, paging files, and software logs. *Safety: Review files manually before deleting. Never delete unfamiliar files in operating system folders.*

4. Academic Disclaimer & Limitations

This report was generated deterministically by the Smart Computer Health Monitor interpretation layer. Correlations between resource metrics and processes are advisory and do not guarantee hardware failure or malware presence. No automated modifications or file deletions have been performed.